



5.7 Further differentiation: implicit and higher derivatives (HL)

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Implicit differentiation is used when y is not isolated as an explicit function of x ; differentiate both sides of the equation with respect to x , treating y as a function of x and applying the chain rule (so $d/dx(y^2) = 2y \cdot dy/dx$), then solve for dy/dx . Higher derivatives (d^2y/dx^2 , etc.) are found by differentiating again, and may require the product/quotient rule if dy/dx itself involves both x and y .

Worked example: follow the reasoning

Find dy/dx for $x^2 + y^2 = 25$ using implicit differentiation.
Differentiate both sides: $2x + 2y(dy/dx) = 0$.
 $dy/dx = -2x/(2y) = -x/y$.

Quick check · multiple choice

Given $y^3 = x$, dy/dx equals:
A $1/(3y^2)$
B $3y^2$
C $1/(3x^2)$
D $3x^2$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider the curve defined implicitly by $x^2 + xy + y^2 = 7$. [8]
(a) Find dy/dx in terms of x and y .
(b) Find the gradient of the curve at the point $(1,2)$.
(c) Find the equation of the tangent to the curve at $(1,2)$.

Final answer: _____



Answers and explanation

Question 1 [8]

- (a) Differentiate term by term: $\frac{d}{dx}(x^2)=2x$; $\frac{d}{dx}(xy)=y+x(\frac{dy}{dx})$ (product rule); $\frac{d}{dx}(y^2)=2y(\frac{dy}{dx})$. So: $2x + y + x(\frac{dy}{dx}) + 2y(\frac{dy}{dx}) = 0 \rightarrow \frac{dy}{dx}(x+2y) = -(2x+y) \rightarrow \frac{dy}{dx} = \frac{-(2x+y)}{(x+2y)}$.
- (b) At (1,2): $\frac{dy}{dx} = \frac{-(2+2)}{(1+4)} = -\frac{4}{5}$.
- (c) Tangent: $y-2 = -\frac{4}{5}(x-1) \rightarrow y = -\frac{4x}{5} + \frac{4}{5} + 2 = -\frac{4x}{5} + \frac{14}{5}$.

Quick check answer

A. Differentiating: $3y^2(\frac{dy}{dx})=1$, so $\frac{dy}{dx}=\frac{1}{(3y^2)}$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.